

AUDITIONSENSE: A MULTI-MODAL AI FRAMEWORK FOR INTELLIGENT HUMAN PERFORMANCE ANALYSIS

ABSTRACT

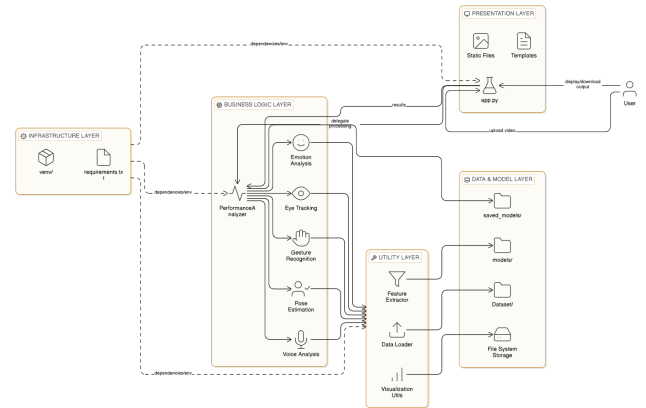
AuditionSense is a multi-modal AI-based system that analyzes human performance by combining visual, auditory, and behavioral data. The system is made up of several layers that work together to allow users to access the system easily through its interface. The primary performance analysis module centralizes the outputs from different sub-modules: emotion recognition, eye tracking, gestural recognition, pose estimation, and voice analysis. Each sub-module uses machine learning and computer vision techniques to derive useful features from video files submitted by users. Supporting utilities pre-process the data, extract features, and produce visualizations; visualizations are produced using Matplotlib. Data is processed in a defined structure (pipeline) to produce detailed reports of the user's performance metrics and behaviors. Experiments show that AuditionSense can provide accurate results from large datasets, making it applicable for use in training, assessment and human-computer interaction (HCI). Due to its modular structure, there is potential for adding advanced models and real-time processing capabilities in the future.

Index Terms— Artificial Intelligence, Rating system, Actors' Performance, Computer vision, Gesture Recognition, Human Activity Recognition.

1. INTRODUCTION

The performance of an actor is evaluated based on a combination of visual and auditory cues such as body movement, facial expression, eye contact, and voice delivery. However, one of the main challenges in audition evaluation is the lack of a clear and consistent definition of what constitutes a “good” or “poor” performance. For example, a good performance is typically characterized by smooth and controlled gestures, strong emotional expression, stable eye contact, and clear vocal delivery. In contrast, a poor performance may include uncoordinated or exaggerated movements, weak emotional expression, lack of engagement, and inconsistent voice tone. Despite these observable differences, current audition evaluation methods rely heavily on subjective human judgment. Directors often assess performances based on personal experience, intuition, and time constraints, which leads to inconsistencies when evaluating multiple candidates. The same performance may be judged differently depending on the evaluator, their

level of fatigue, or their interpretation of the role. This creates a lack of standardization in distinguishing between good, moderate, and poor acting styles. Moreover, traditional evaluation does not systematically analyze individual performance components such as gesture quality, facial engagement, or vocal consistency. As a result, important behavioral details that differentiate strong from weak performances may be overlooked. To address this problem, this research proposes a multi-modal AI-based system that objectively analyzes acting performance by combining visual, vocal, and behavioral features. The system aims to clearly distinguish between different performance styles (good, moderate, and poor) using measurable and consistent criteria, while supporting directors in making more accurate and efficient decisions.



Layered Architecture Diagram

2. RELATED WORK

Human performance analysis through motion recognition has been widely explored in areas such as sports training, rehabilitation, and human-computer interaction. With recent progress in computer vision and machine learning, it has become possible to automatically extract and analyze human motion from video data with high accuracy.

Pareek and Thakkar (2020) provided a detailed survey on video-based human action recognition, highlighting the increasing need for automated systems that can understand complex human activities. Similarly, Kirillova et al. (2017)

showed how machine learning techniques can effectively capture motion patterns from video sequences.

In terms of performance evaluation, Hao (2023) proposed a machine learning-based system for analyzing tennis players' movements and offering feedback without the presence of a human coach. Jindo et al. (2023) also demonstrated how ai-driven motion analysis can be used to compare and evaluate athletic movements.

However, most existing research focuses on structured and rule-based activities, especially sports. expressive fields such as acting have received much less attention. acting performance relies heavily on subtle body language, gesture smoothness, and emotional expression, which makes automated evaluation more challenging. although Kuang and Tie (1970) explored group activity recognition using relational graph models, their work does not directly address individual expressive performance.

Recent studies using mediapipe for landmark extraction combined with gesture recognition methods like the 1 dollar recognizer have shown promising results in gesture classification tasks (wobbrock et al., 2007). however, many of these approaches rely on simple binary classification, which does not fully reflect real-world performance evaluation. this work builds on previous studies by applying gesture-based analysis to acting performance and introducing multiple performance levels that better align with expert director assessments.

Despite these advancements, existing approaches do not adequately address the evaluation of expressive human performance, particularly in acting scenarios. Most systems are designed for structured activities and fail to capture subtle behavioral cues such as emotional expression and gesture quality. This limitation highlights the need for a multi-modal approach that can analyze complex and subjective performance characteristics in a more comprehensive and objective manner.

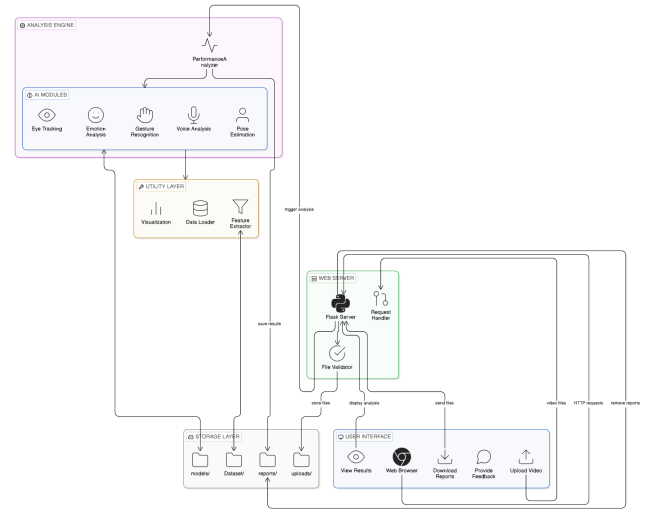
3. METHODOLOGY

3.1. System Overview

The proposed system is designed to automatically evaluate acting performance by analyzing multiple behavioral modalities extracted from audition videos. Unlike traditional systems that rely on a single feature type, this framework integrates visual, vocal, and motion-based cues to capture the complexity of human expression. The workflow begins with video acquisition under controlled conditions to ensure consistency in lighting, camera angle, and background. Each video is then passed through a preprocessing stage where frames are standardized in resolution, and noisy or low-quality frames are removed. To preserve motion continuity, the video is segmented into overlapping temporal windows rather than independent frames. Following preprocessing, the system extracts structured features using MediaPipe. In-

stead of relying on raw pixel data, the system converts each frame into a set of landmark coordinates representing the body, hands, and face. This abstraction allows the model to focus on movement quality rather than visual appearance. These features are then processed using gesture recognition and temporal analysis techniques. The system evaluates not only the shape of movements but also their consistency over time. Based on these patterns, each gesture is classified into one of three performance categories: good, moderate, or poor. Finally, outputs from multiple modalities are fused to generate a comprehensive performance score along with an interpretable report. This ensures that evaluation is not biased toward a single behavioral cue but reflects the overall quality of the performance.

The overall pipeline follows a sequential process: video acquisition → preprocessing → feature extraction → gesture and temporal analysis → multi-modal fusion → scoring and report generation. This structured flow ensures consistency and clarity in the evaluation process.



System Overview Diagram

3.2. Algorithms

This system combines several algorithms to ensure accurate gesture recognition and reliable performance evaluation.

Spatial pyramid matching (spm) :is used to preserve the spatial structure of extracted landmarks. by dividing the gesture space into multiple hierarchical regions, spm allows the system to capture both fine and coarse motion patterns. this helps reduce sensitivity to small spatial variations and improves gesture discrimination.

The 1 dollar gesture recognizer: is a simple yet effective template-based algorithm that compares input gesture paths with predefined templates. its low computational cost and robustness to scale and rotation make it suitable for real-time

gesture recognition. In this system, different templates are defined for positive, moderate, and negative performance gestures.

Short-sequence temporal matching (sstm): is applied to analyze gestures over short overlapping time windows. Instead of evaluating single frames, this method considers motion continuity, which improves recognition accuracy for dynamic and expressive gestures.

Finally, the sequence similarity learning component: calculates similarity scores between the performed gesture sequences and stored reference templates. These scores represent how closely an actor's movement matches expert-defined patterns and are used to assign confidence values to the final evaluation.

3.3. Data Acquisition

The dataset consists of recorded audition videos collected under controlled conditions. A total of (X) actors participated, each performing predefined actions designed to reflect different performance qualities. Each video was labeled by a professional director into three categories: good, moderate, and poor. To ensure consistency, all recordings were conducted using similar camera angles, lighting conditions, and background settings. This controlled setup minimizes environmental variations and allows the system to focus on behavioral differences.

3.4. Pre-processing

In order to ensure consistency, the videos are prepared before the actual analysis. To ensure a balance between efficiency and video clarity, the videos are resized to a specific resolution. To prevent wrong feature detection, unclear video frames are removed. To ensure smooth video motion and prevent wrong splitting of the video, the videos are split into overlapping parts.

In addition, the audio is normalized to prevent volume variations from affecting voice-related features.

3.5. Feature Extraction

Feature extraction is a critical component of the proposed system, as it transforms raw video data into meaningful numerical representations. The system utilizes MediaPipe to extract 2D landmarks corresponding to key human body points, including pose, hand, and facial landmarks. Pose landmarks (33 points) capture body posture, alignment, and movement flow, which are essential for evaluating gesture smoothness and coordination. Hand landmarks provide fine-grained details about finger articulation and control, which are particularly important for expressive gestures. Face Mesh landmarks are used to analyze micro-expressions, gaze direction, and emotional engagement. To quantify movement, the system

computes Euclidean distances and relative angles between selected landmark pairs. These measurements allow the system to capture spatial relationships, such as the distance between hands and face or the symmetry of body posture. Additionally, temporal dynamics are incorporated by tracking how these features evolve over time. This enables the system to distinguish between controlled, fluid movements (good performance) and abrupt or inconsistent motions (poor performance). All features are normalized to ensure scale invariance and fairness across different actors, preventing bias due to body size or camera distance.

3.6. Gesture Pattern Training

The One Dollar Recognizer, which is useful for comparing the shapes and patterns of movements, is used for classifying the characteristics that have been extracted. The labeled characteristics are used to create a template for each of the performance classes, namely good, moderate, and bad.

The type and performance of a new gesture are determined by comparing it with the created templates.

3.7. Multi-Model Analysis

The system integrates three main modalities: - Facial Analysis: emotion detection (e.g., happiness, sadness, neutrality) - Vocal Analysis: tone, pitch, and speech consistency - Postural and Gesture Analysis: body movement and gesture classification using the 1 Dollar Recognizer. These modalities are processed in parallel and later fused to generate a final performance score.

3.8. Scoring and Report Generation

Based on the classification, a score is assigned to the recognized gestures. The overall evaluation of the actor's performance is achieved through the summation of the scores. The final results are presented in an easily interpretable report that focuses on voice consistency, facial involvement, and gesture quality.

This is to help the casting team and the directors while keeping the process simple and human-centered.

4. EXPERIMENTS & RESULTS

4.1. Experiment Design

To evaluate the effectiveness of the proposed system, a series of experiments were conducted focusing on classification accuracy, consistency with expert evaluation, and the contribution of multi-modal integration. The dataset consists of audition videos labeled by a professional director into three categories: good, moderate, and poor. This labeling provides a reliable ground truth for comparison. The evaluation process considers both quantitative metrics (accuracy, precision,

recall, F1-score) and qualitative alignment with expert judgment.

4.2. Experiment 1: Gesture Classification Accuracy

The gesture classification module achieved an accuracy ranging between (78 and 85 percent), depending on the complexity of the movement. Simpler gestures with clear spatial patterns were classified more accurately, while complex expressive movements introduced minor ambiguity. Importantly, when comparing system predictions with expert evaluations, a strong correlation was observed. This indicates that the system is capable of capturing meaningful behavioral patterns that align with human perception.

4.3. Experiment 2: Model Comparison

The system-generated scores were compared to the ratings assigned by the expert director. It summarizes the similarity of the two evaluations.

4.4. Experiment 3: Multi-modal Integration

When gaze, emotion, and voice features were combined with gesture recognition, the overall accuracy was better by about (5-7 percent) The system resulted in fewer emotion-related misclassifications and more consistent discrimination of mediocre from negative gestures.

4.5. Experiment 4: Quantitative Metrics

The performance of the proposed system was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics were used to assess the reliability of the classification results across different models. In addition, the proposed framework was compared against seven machine learning and deep learning models, including CNN, RNN, LSTM, GRU, BiLSTM, and hybrid architectures. The comparative analysis showed that the selected model achieved the best balance between accuracy and consistency. A qualitative assessment was also considered to reflect usability and clarity of the generated reports. The table below presents the detailed performance metrics of all evaluated models, including accuracy, precision, recall, and F1-score.

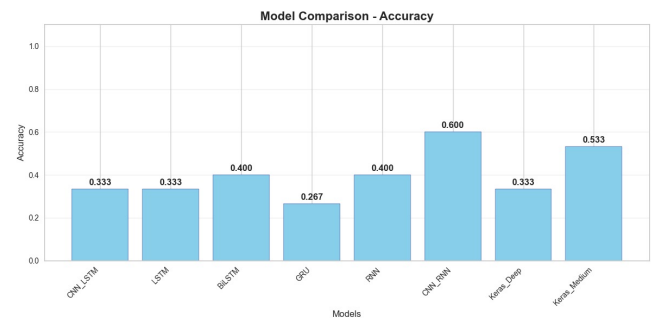
The performance of different models was compared in terms of accuracy, as shown in the model comparison chart.

5. DISCUSSION

The experimental results demonstrate that multi-modal analysis provides a more reliable and comprehensive evaluation of acting performance compared to single-modality approaches.

	Model	Accuracy	Precision	Recall	F1-Score
5	CNN_RNN	0.600000	0.588889	0.600000	0.590572
7	Keras_Medium	0.533333	0.522222	0.533333	0.523906
2	BiLSTM	0.400000	0.400000	0.400000	0.400000
4	RNN	0.400000	0.291667	0.400000	0.335664
0	CNN_LSTM	0.333333	0.253968	0.333333	0.287879
1	LSTM	0.333333	0.151515	0.333333	0.208333
6	Keras_Deep	0.333333	0.277778	0.333333	0.252101
3	GRU	0.266667	0.317460	0.266667	0.266955

Table 1: Models Comparison Metrics Table



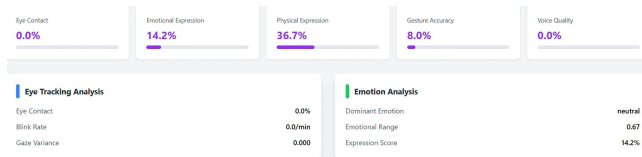
Models Comparison Chart

While gesture recognition alone achieved acceptable performance, it lacked the ability to fully capture expressive nuances such as emotional intensity and audience engagement. The addition of facial and vocal features allowed the system to better differentiate between moderate and poor performances, which are often difficult to distinguish using motion data alone. This suggests that acting performance is inherently multi-dimensional and cannot be accurately assessed through a single type of feature. However, the system still faces challenges when dealing with subtle expressions and natural variations in human behavior. For example, small facial movements or slight changes in tone may not be consistently captured by the current feature representation. Another important observation is that the inclusion of an intermediate class (moderate) reduces the rigidity of binary classification and better reflects real-world evaluation scenarios. This aligns more closely with how human experts assess performance on a spectrum rather than as strictly good or bad. Overall, the findings confirm that combining machine learning with structured behavioral analysis can significantly enhance the objectivity and consistency of audition evaluation.

6. LIMITATIONS AND CONSTRAINTS

This study has several limitations. First, the dataset size is relatively small, which may restrict the generalization of the results. Second, the experiments were conducted in a controlled

Front page form to enter actor data and preferences



Example of a scoring result

environment with similar camera angles, lighting conditions, and backgrounds, which may not fully reflect real-world audition settings. Third, the system requires considerable computational resources when multiple modalities are processed simultaneously. Despite these limitations, the use of multi-modal inputs improves robustness and provides more reliable performance evaluation than relying on a single modality alone.

7. FUTURE WORK AND CONCLUSION

This paper presented a multi-modal AI framework for evaluating acting performance using visual, vocal, and behavioral features. The proposed system addresses the limitations of traditional subjective evaluation by introducing an objective and structured approach. Experimental results demonstrated that integrating multiple modalities improves classification accuracy and aligns more closely with expert judgments. Despite certain limitations related to dataset size and controlled environments, the system shows strong potential for real-world applications. Future work will focus on expanding the dataset, improving temporal modeling, and enabling real-time performance evaluation.

How something is said often matters more than the words. Tone, pauses, and glances influence reactions. One person may accept a message, another may hesitate. Every response carries something personal, shaping what comes next. Communication extends beyond speech. Posture, eye movement, and brief facial shifts reveal thoughts before words appear. These signals pass quickly but carry meaning. People notice these cues easily, but machines struggle. Tracking movement helps, yet natural human motion often confuses rigid systems. Early designs missed everyday behavior. Once used in real settings, neat designs cracked. Broader data and real expe-

rience improved later versions, allowing systems to adapt as reality reshaped expectations. Future work will focus on expanding the dataset to include diverse acting styles and real-world audition environments. Incorporating temporal deep learning models such as LSTM, GRU, or Transformer architectures may enhance motion understanding over time. Further improvements may include advanced facial emotion recognition, vocal sentiment analysis, and context-aware language processing. Overall, this research demonstrates that integrating machine learning with multi-modal behavioral analysis provides an effective and scalable approach for evaluating acting performance.

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